UNIVERSITY OF BAGHDAD

COLLEGE OF MEDICINE

2026-2027

Imaging of pulmonary neoplasma

عنوان المحاضرة

DEPARTMENT OF SURGERY / RADIOLOGY BRANCH

Stage four • Academic Year 2026–2027

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Learning Objectives

- Recognize the classification of chest neoplasms
- Compare between radiological sings malignant and benign pulmonary masses
- Identify common benign and malignant pulmonary neoplasm.
- Recognize high-yield points relevant to assessment.

Classification of Chest Neoplasm

- Tumors of the airways.
- Lung tumors .
- Mediastinal tumors .
- Pleural tumors.
- Chest wall tumors

Lung Tumors

Malignant

- Bronchogenic carcinoma
- Lymphoma
- Metastasis
- Sarcoma (rare)

Low grade malignancies

- Carcinoid tumor

Benign (rare)

- hamartoma

Benign pulmonary neoplasms

Benign tumors, rare and most commonly being hamartoma

Hamartoma is the most common benign tumor of the lung.

composed of cartilage (predominant component), connective tissue, muscle, fat, and bone

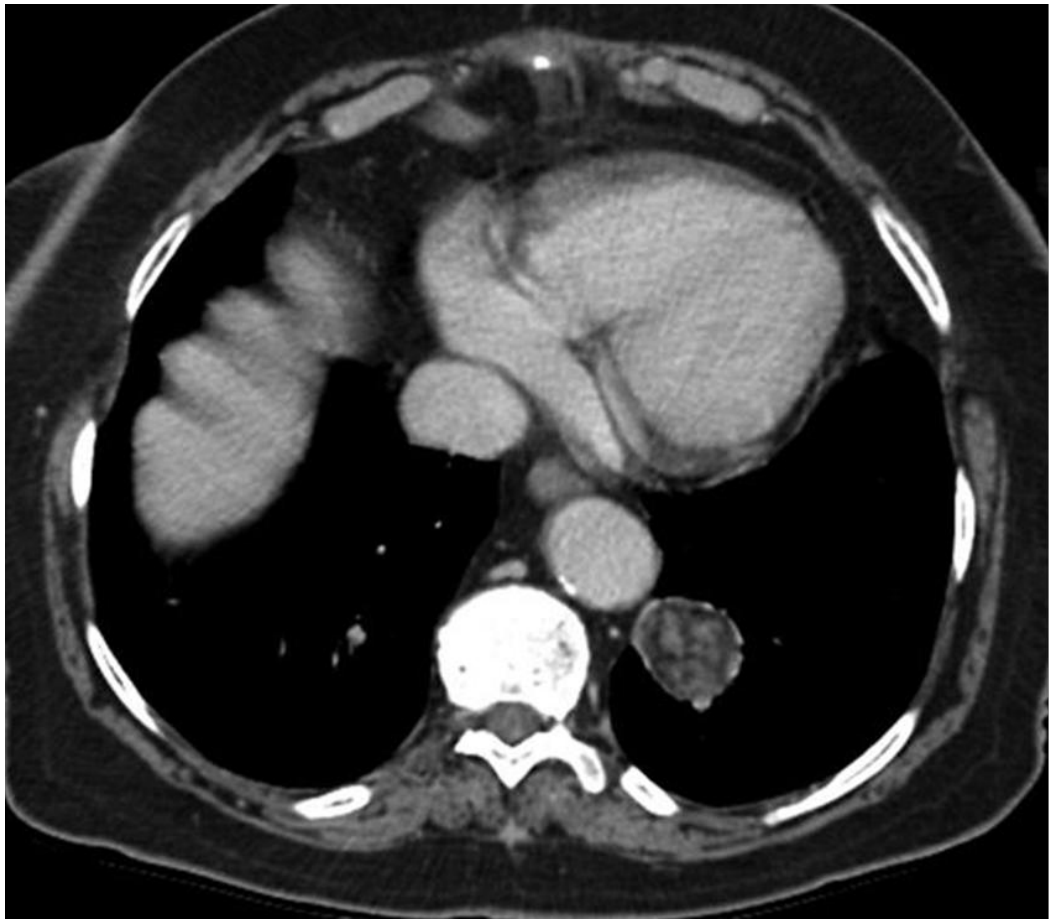
90% are peripheral, 10% are endobronchial

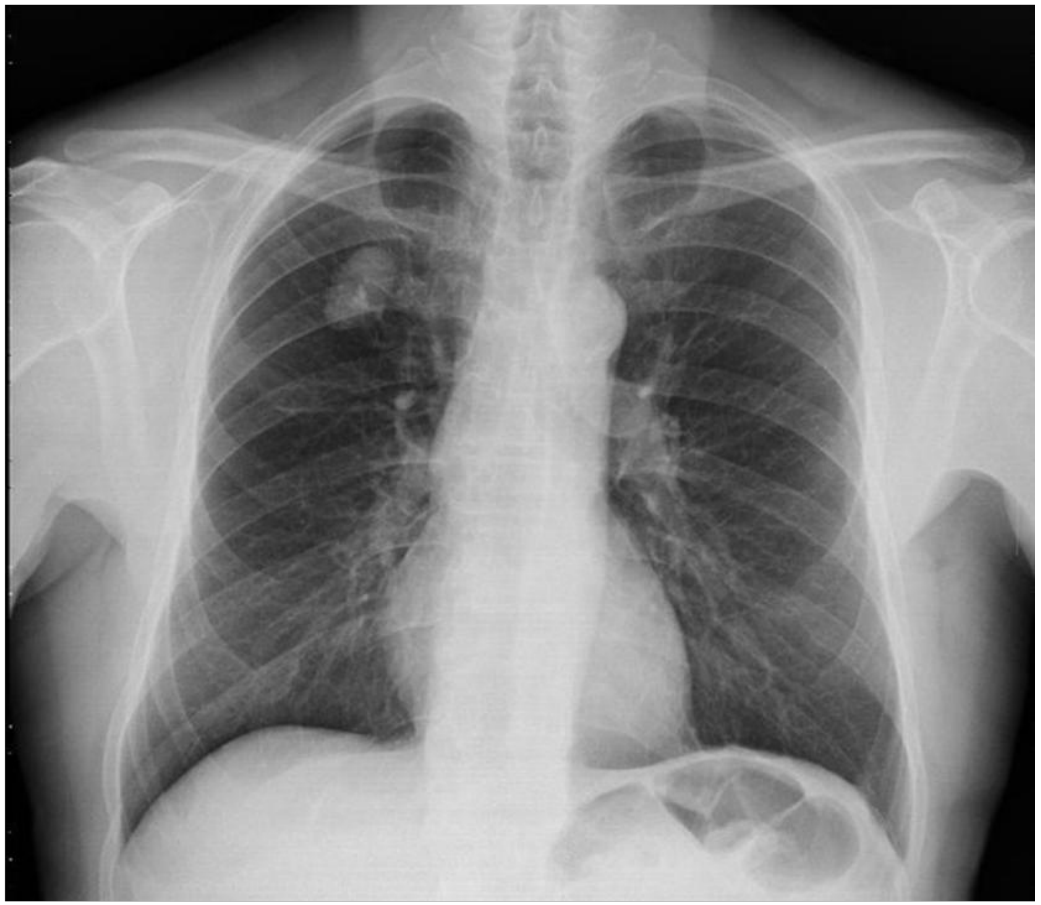
Usually asymptomatic and detected incidentally.

Radiographic Features:

- Well-circumscribed solitary nodule.
- Chondroid "popcorn" calcification is diagnostic but uncommon (20%)
- Fat attenuation within a lesion by CT is pathognomonic.

Benign pulmonary neoplasms





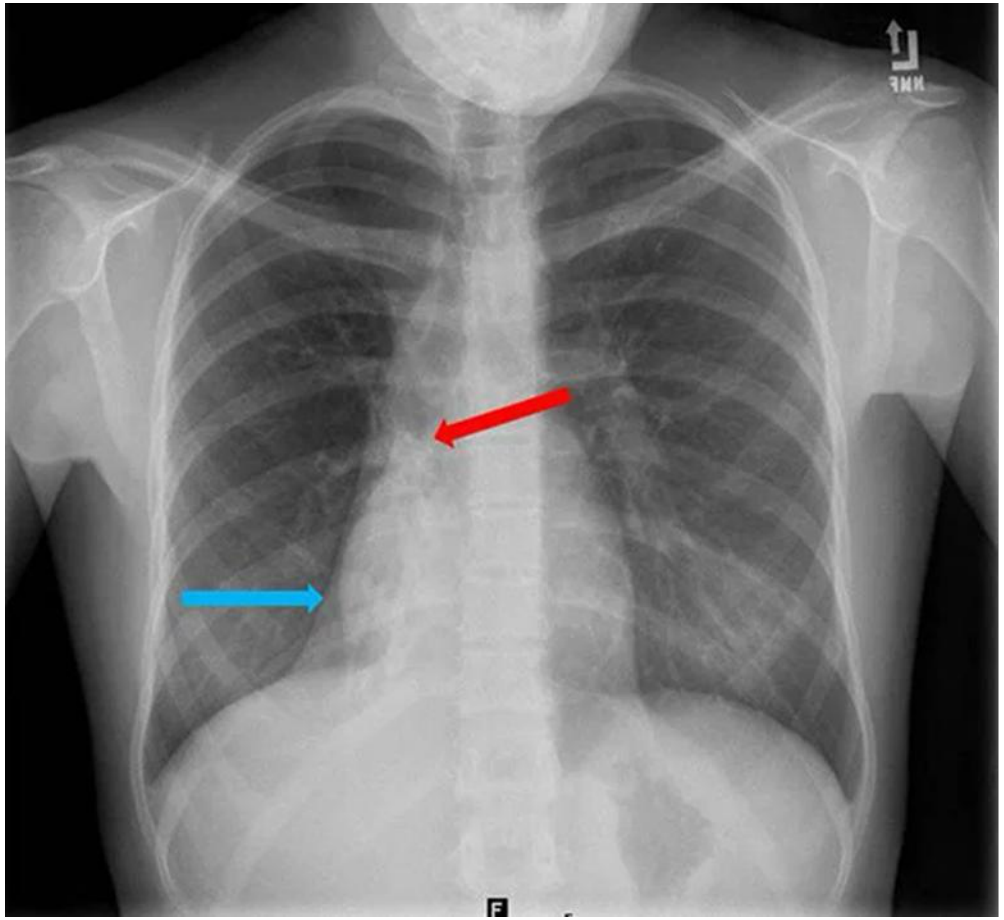
Pulmonary Carcinoid Tumor

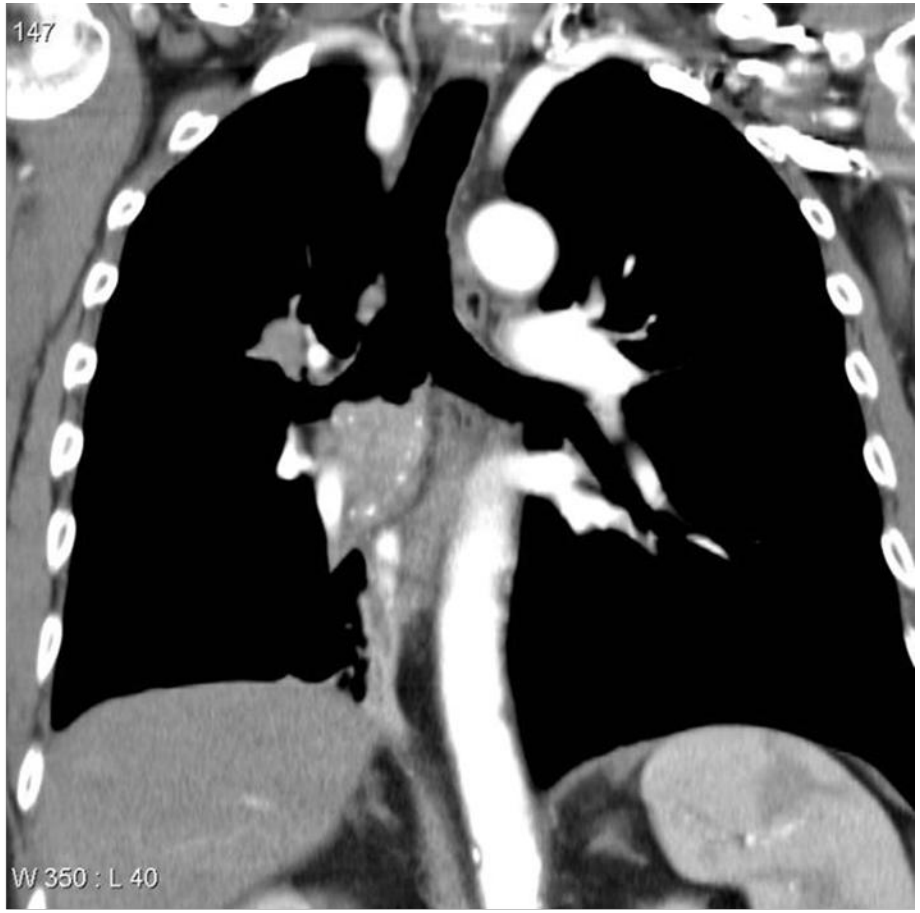
- Is a subgroup of neuroendocrine tumors of the lung
- Central lesions (bronchial carcinoid) commoner (~ 75%) than peripheral pulmonary carcinoid.
- Typical carcinoid , which is well differentiated and low grade is more common than atypical carcinoid.

Radiological features (Central lesions):

- A single hilar or perihilar mass which is usually well-defined, round or ovoid.
- Typically, the size (2-5 cm).
- Marked homogeneous contrast enhancement due to high vascularity.
- Calcification (usually eccentric) can occur but is not a common feature

Pulmonary Carcinoid Tumor





Bronchogenic Carcinoma

Non-small cell lung cancer (80%)

- Adenocarcinoma (most common), 35-40 %, peripheral, more in Females and in non-smokers but still seen in smokers.
- Squamous cell carcinoma, 30%, +++++association with smoking, cavitate, poor prognosis.
- Large cell carcinoma, 15%, peripheral, large in size

Small cell lung cancer (15- 20%)

- Associated with smoking, rapidly disseminated, superior vena cava obstruction, poor prognosis.

Bronchogenic carcinoma

Risk factors

- Smoking (strongest relationship between smoking and cancer has been established for SCC, followed by adenocarcinoma).
- Radiation, uranium mining.
- Asbestos exposure.
- Genetic predisposition (HLA-Bw44 associated?).

Bronchogenic carcinoma

LOCATIONS OF TUMORS

Tumor	Frequency (%)	Location	Comments
Adenocarcinoma	40	Peripheral	Scar carcinoma
Squamous cell carcinoma	30	Central, peripheral[a]	Cavitation
Small cell carcinoma	15	Central, peripheral[a]	Endocrine activity
Large cell carcinoma	1	Central, peripheral	Large mass

[a]Rarely present as a resectable T1 lesion.

Radiological signs

Primary signs

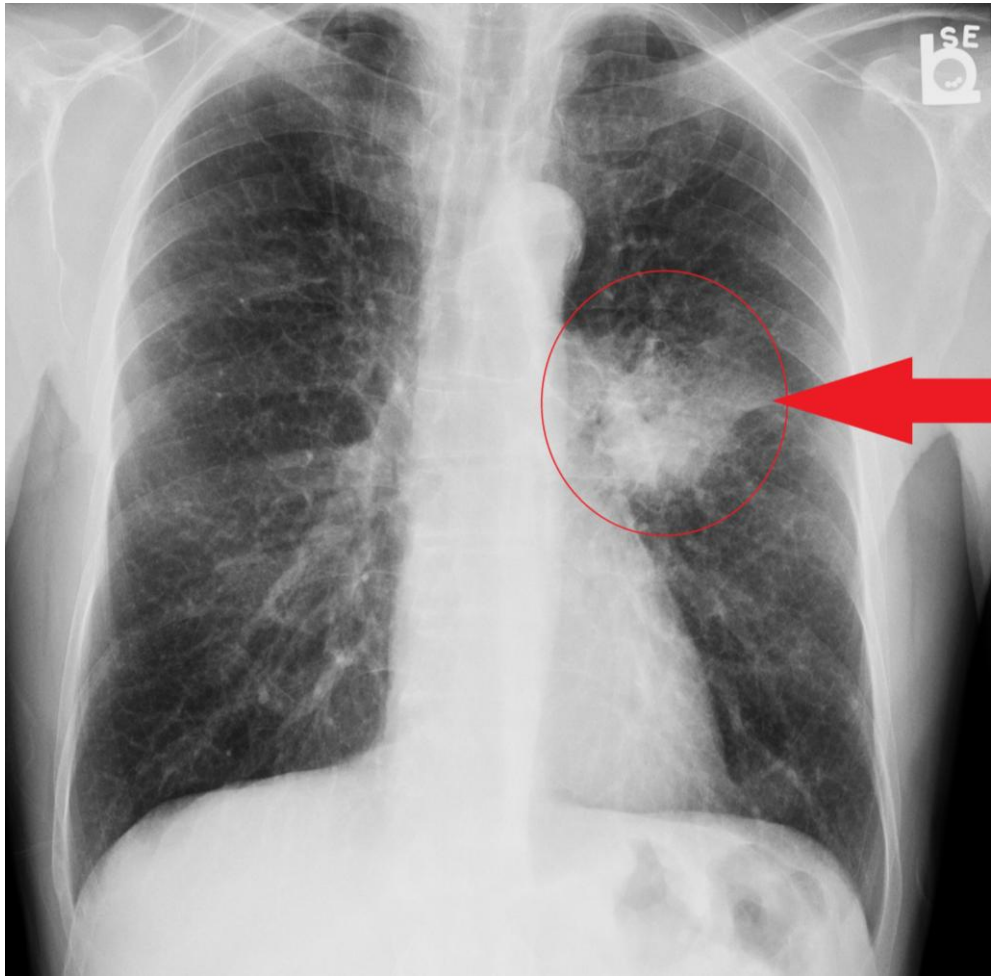
- Mass > 6 cm
- Speculated / irregular margin
- Cavitation (> 15 mm wall thickness is suggestive of bronchogenic CA).
- Unilateral enlargement of hilum (mediastinal widening, hilar prominence).

Secondary signs

- Atelectasis (Golden inverted S sign in RUL, LUL collapse).
- Obstructive pneumonia.
- Pleural effusion.
- Interstitial patterns: lymphangitic tumor spread.
- Hilar and mediastinal adenopathy.
- Metastases to ipsilateral or contralateral lung.

Radiological signs

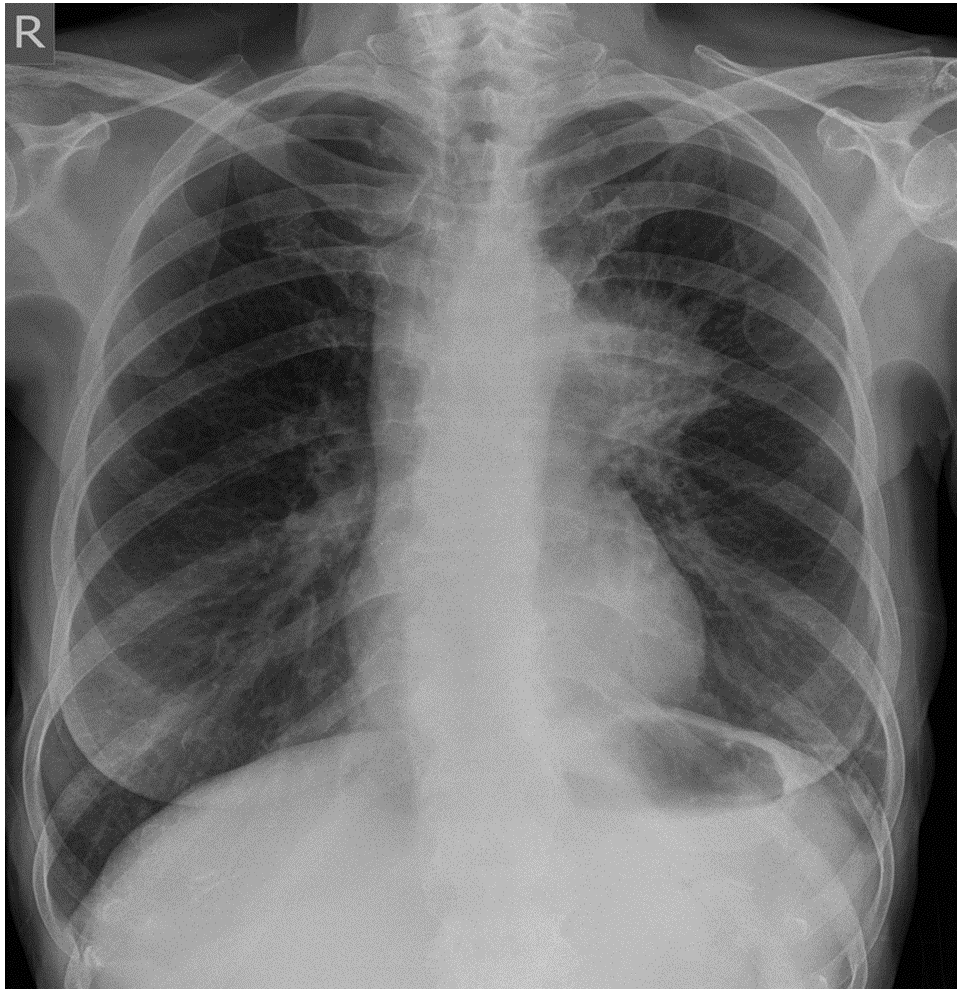


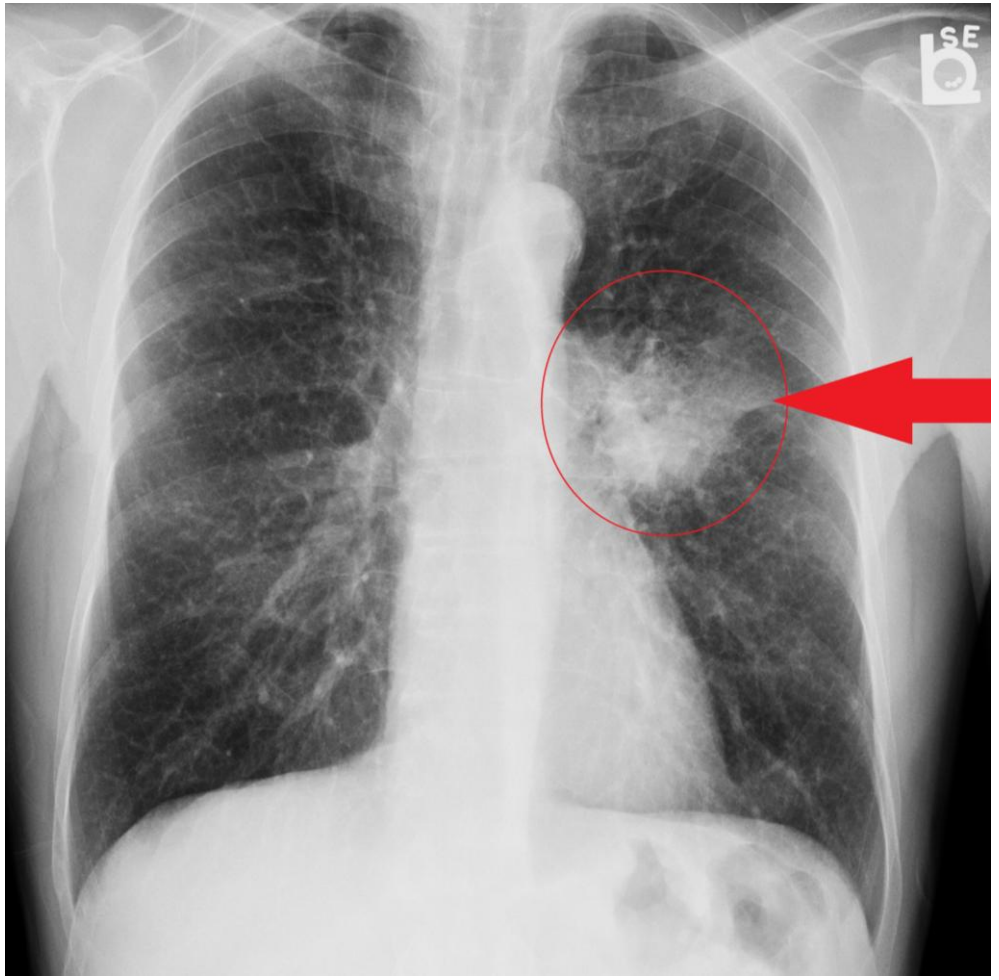


Radiological signs (mass)



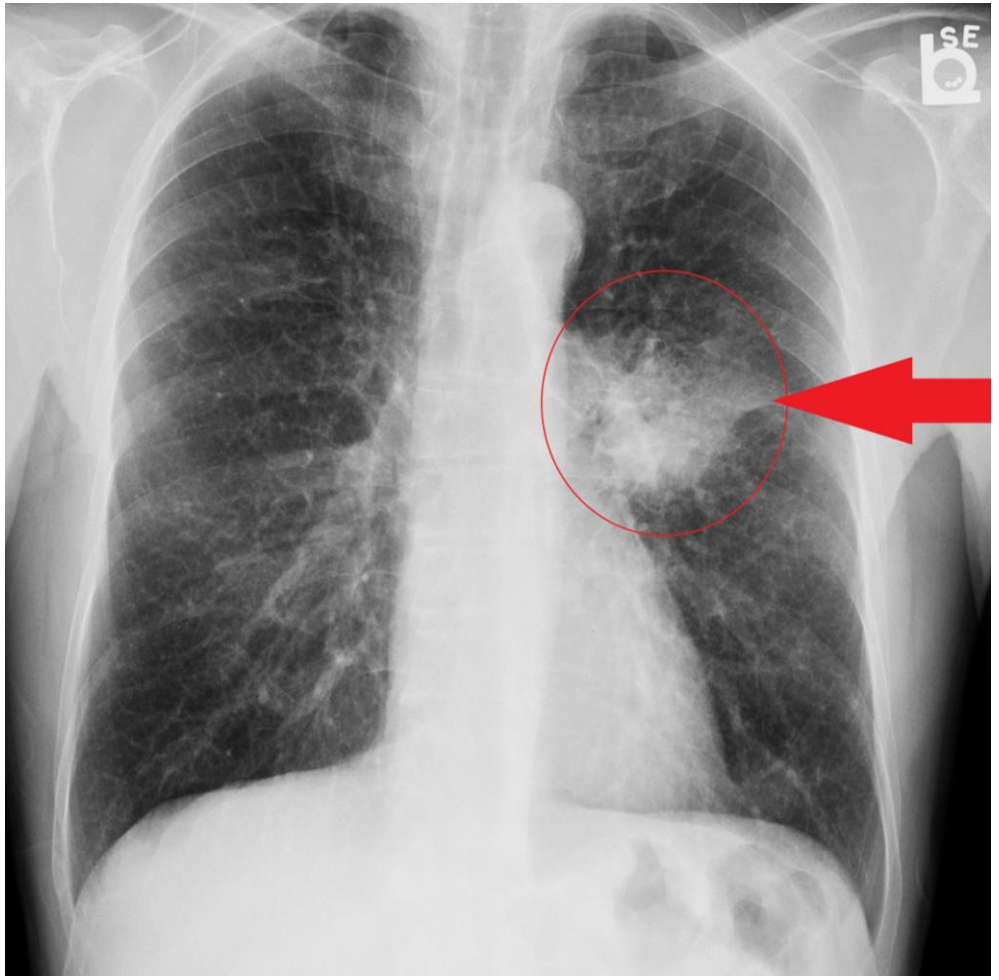
Radiological signs (speculated margin)



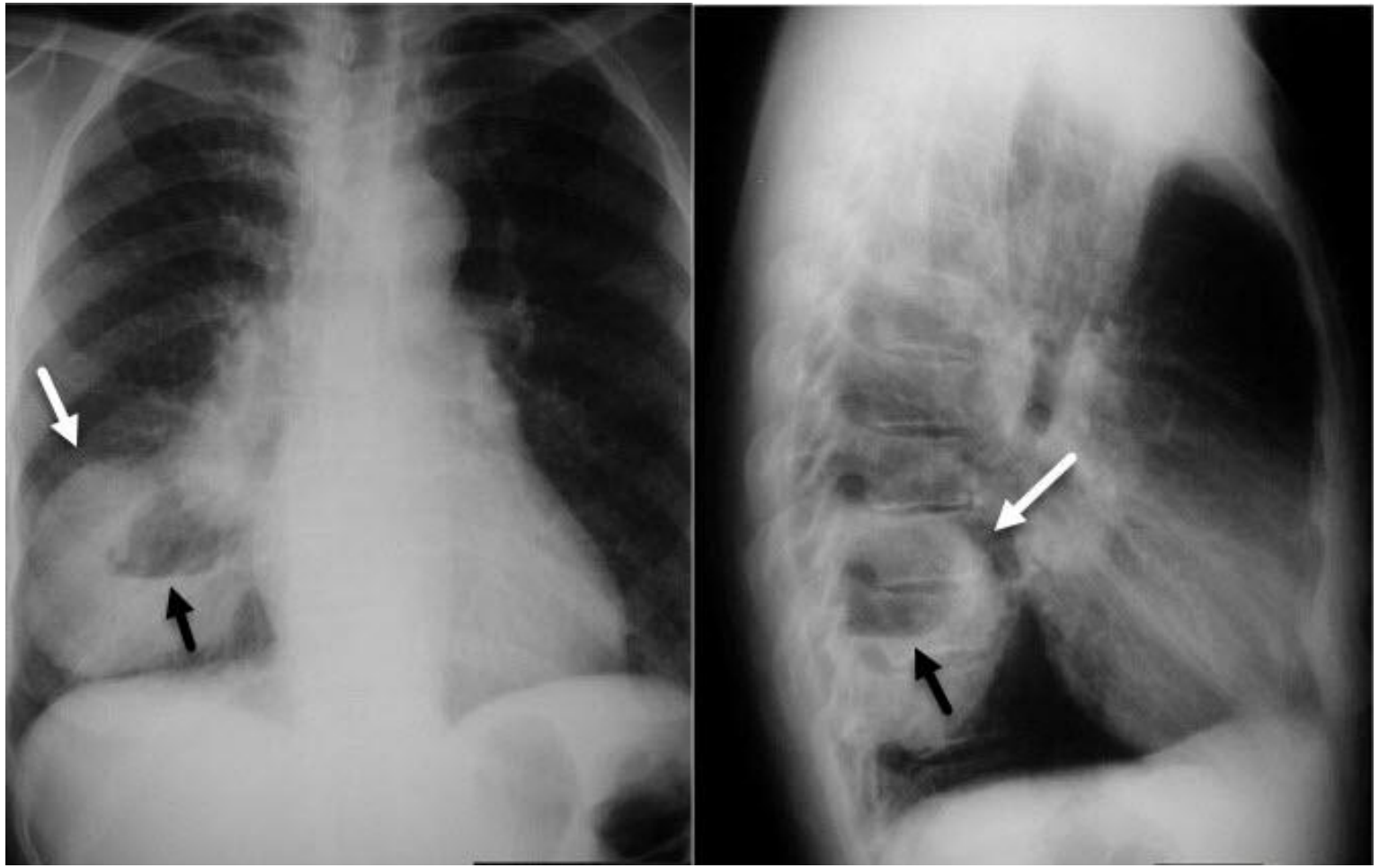


Radiological signs

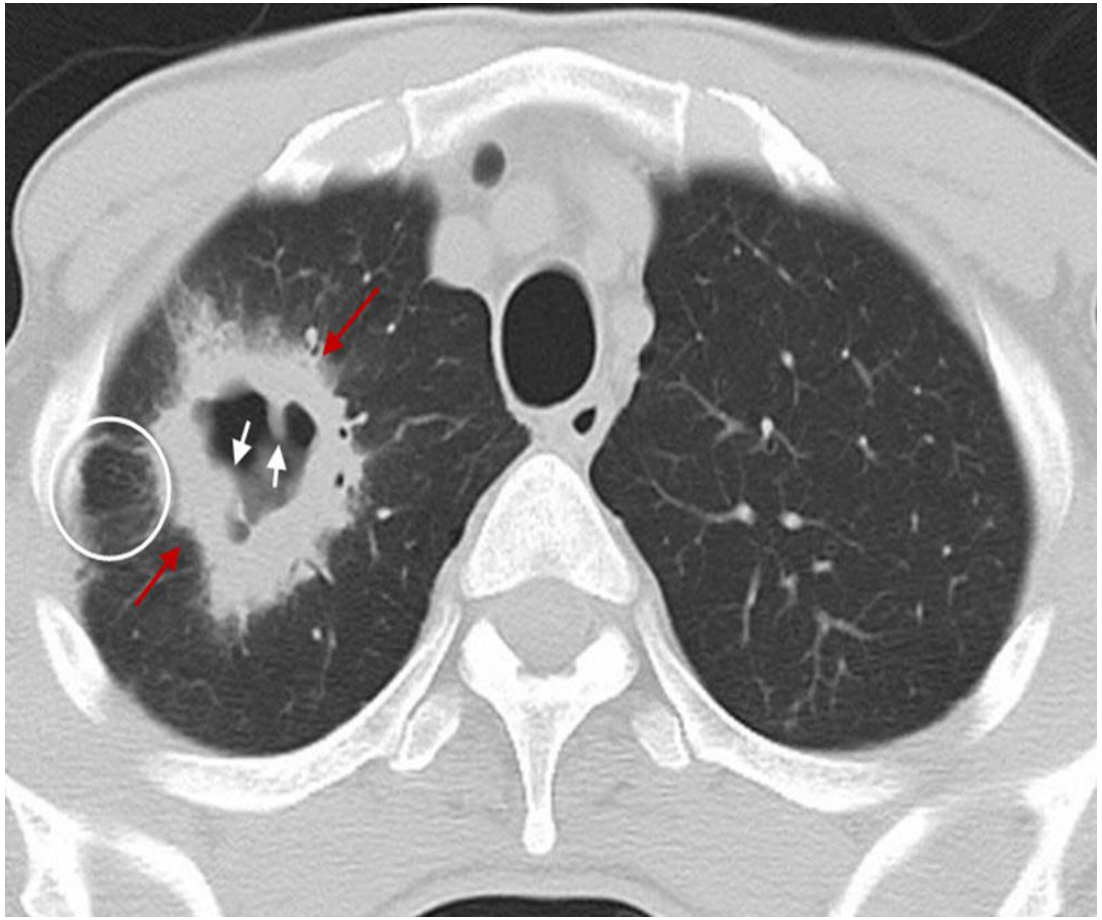




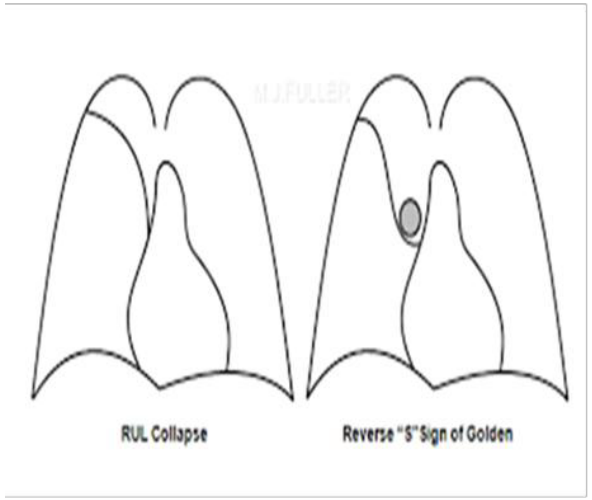
Radiological signs (Cavitation)

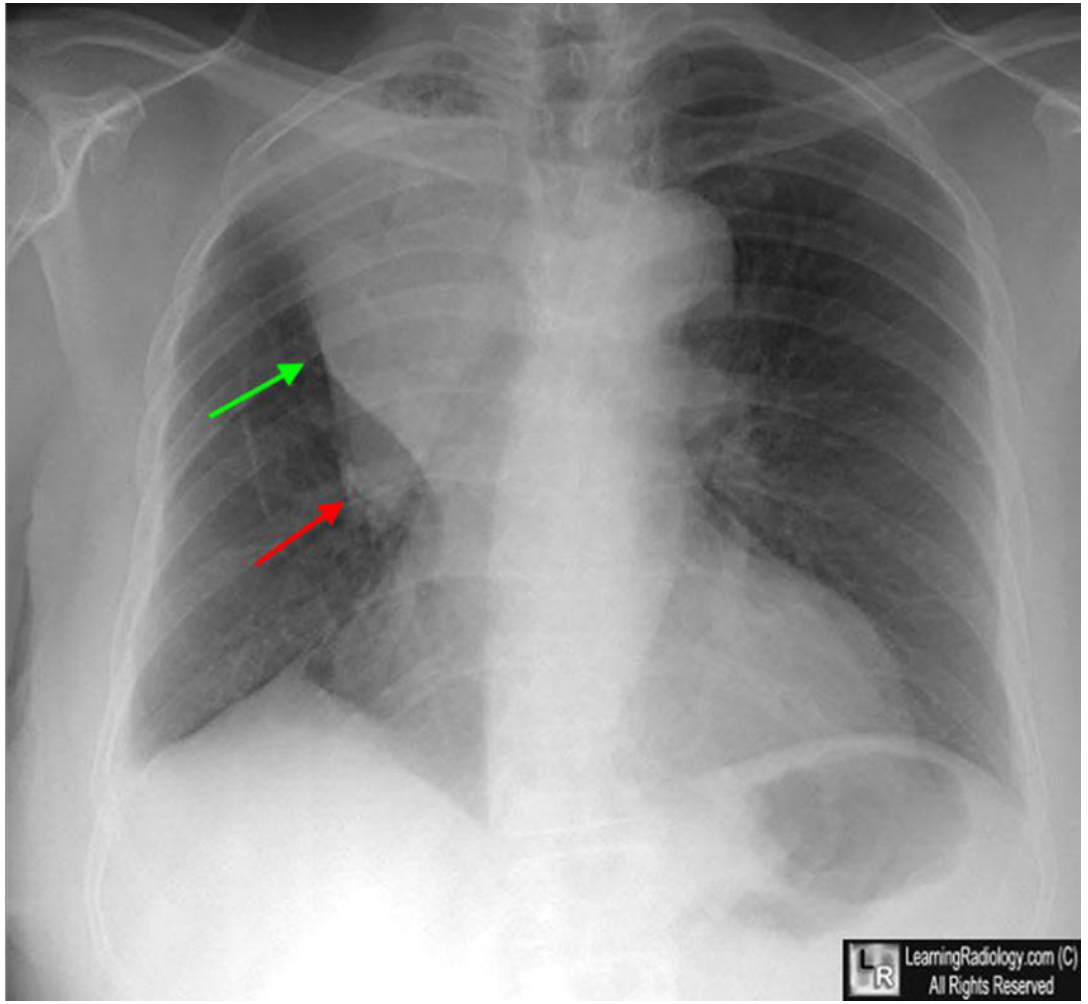


Radiological signs (Cavitation)

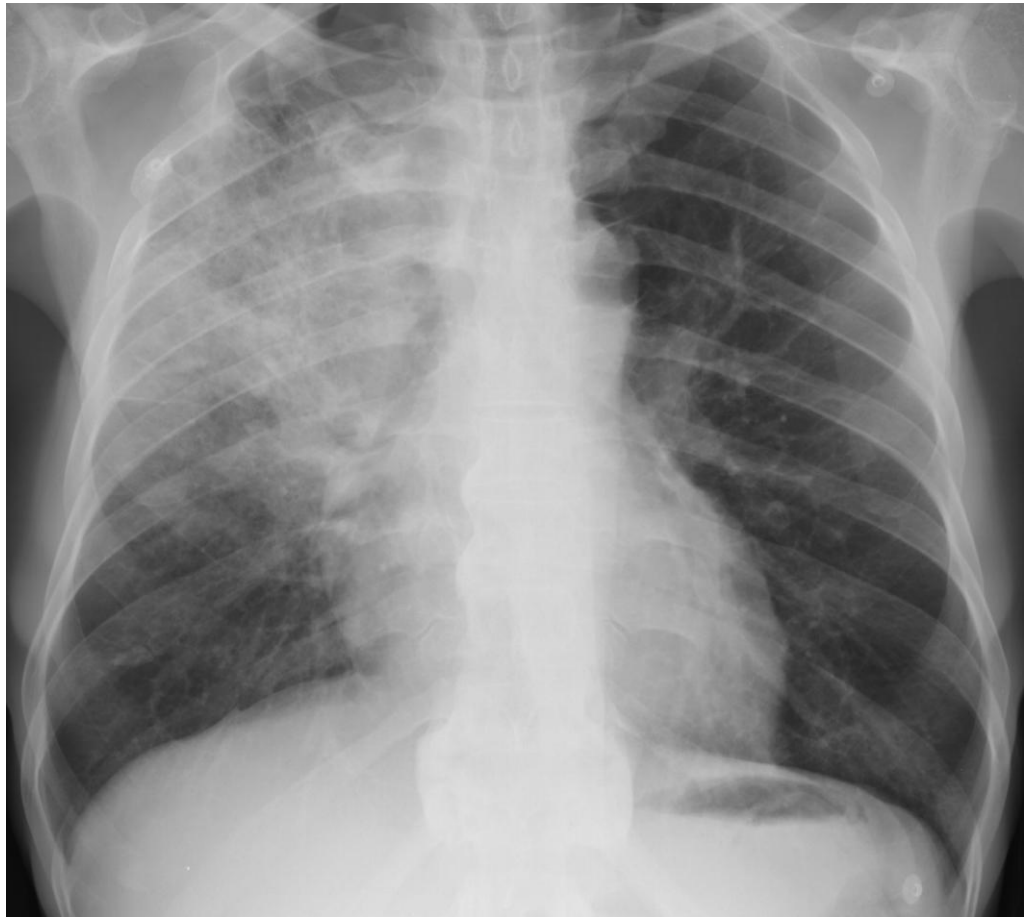


Radiological signs (Golden S sign)





Obstructive pneumonia.



TNM Staging System, (Non-small Cell Lung Cancer)

Tumor (T)

T0 : No evidence of a primary tumor.

T1: ≤3 cm, limited to lung, not more proximal than lobar bronchus.

T2: >3 but ≤7 cm

- Or invades parietal pleura.
- Or involves main bronchus ≥2 cm distal to carina.
- Or atelectasis/obstructive pneumonia extending to hilum but not involving entire lung.

T3: >7 cm

- Or directly invades chest wall, diaphragm, phrenic nerve, mediastinal pleura, parietal pericardium
- Or mainstem bronchus <2 cm distal to carina.
- Or atelectasis/obstructive pneumonia involving entire lung.
- Or separate tumor nodules in a same ipsilateral lobe.

T4: Any size invading heart, mediastinum, great vessels, trachea, recurrent laryngeal nerve, esophagus, vertebral body, or carina

- Or separate tumor nodules in a different ipsilateral lobe

TNM Staging System, (Non-small Cell Lung Cancer)

Nodes (N)

N 0 : no lymph nodes.

N1 : ipsilateral hilar lymph nodes.

N2 : ipsilateral mediastinal lymph nodes.

N3 : contra lateral hilar or mediastinal

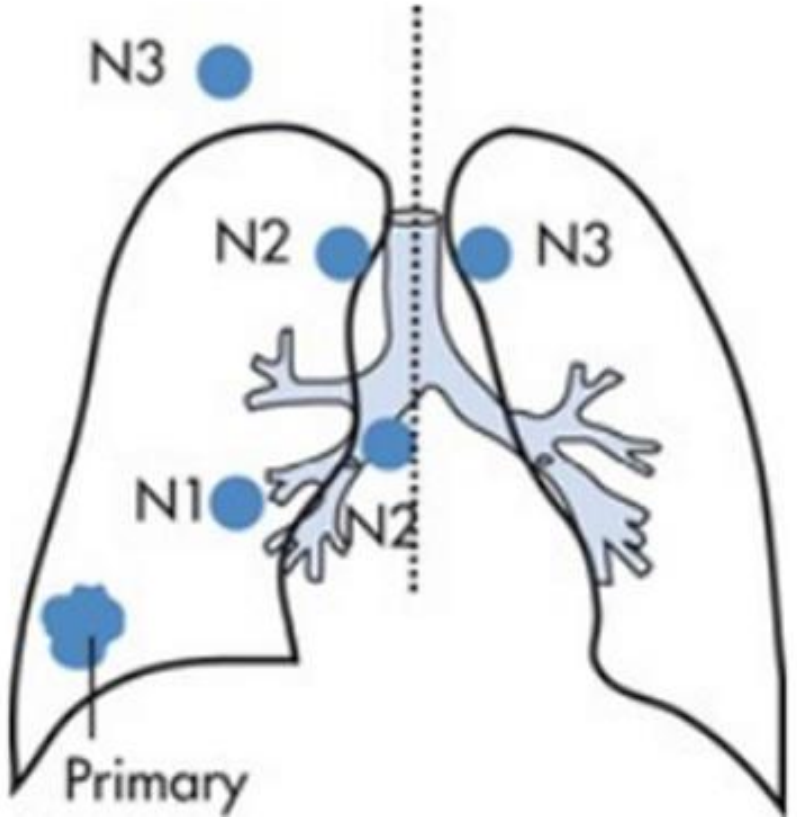
lymph nodes, ipsilateral or contra lateral supra clavicular or scalene lymph nodes.

Metastasis (M)

M 0 : no metastasis.

M1 : metastasis:

- M1-a intra thoracic
- M1-b extra thoracic



Small Cell Lung Cancer

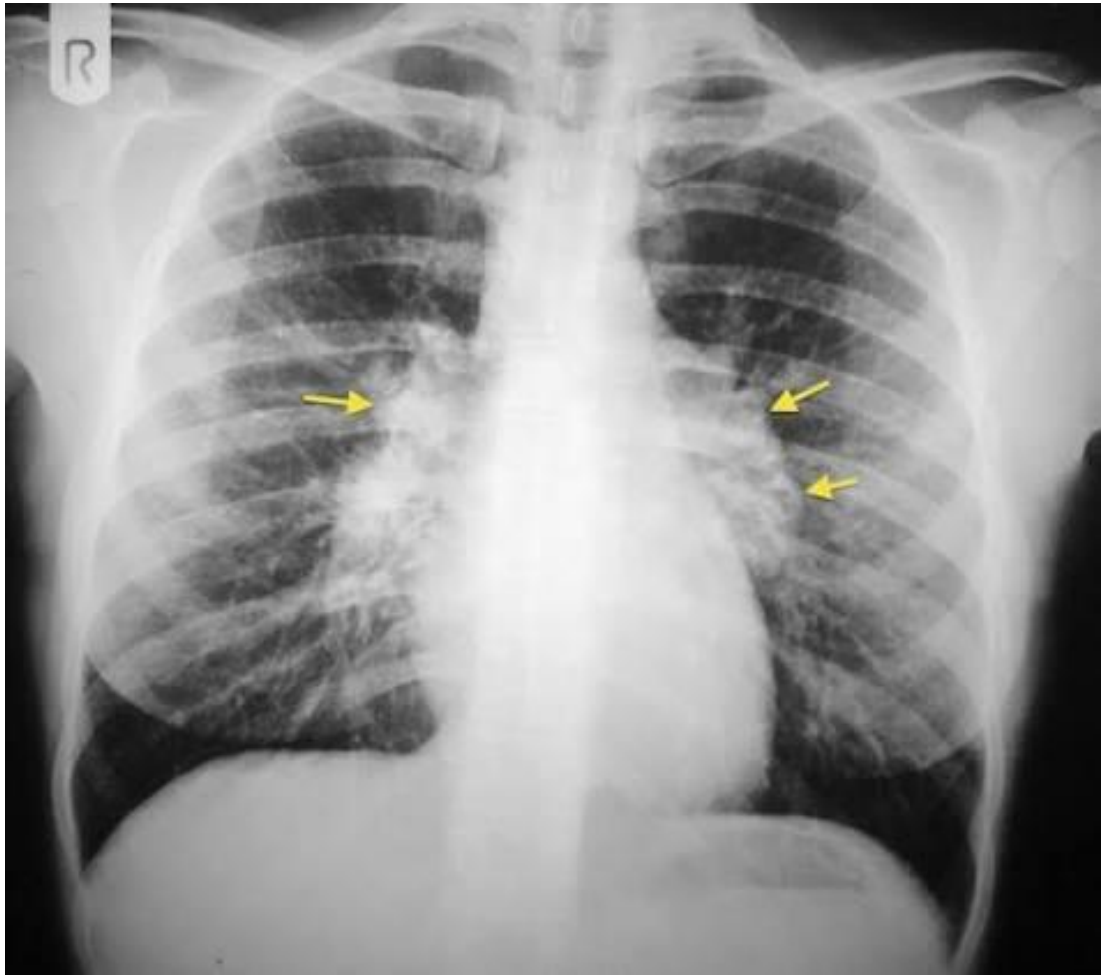
- Accounts for 15% of all lung cancers.
- Most aggressive lung tumor with poor prognosis.
- At diagnosis, two thirds of patients already have extrathoracic spread:
- Typical initial presentation: massive bilateral lymphadenopathy
- With or without lobar collapse
- Brain metastases.

◈ Two-stage system had been used for SCLC:

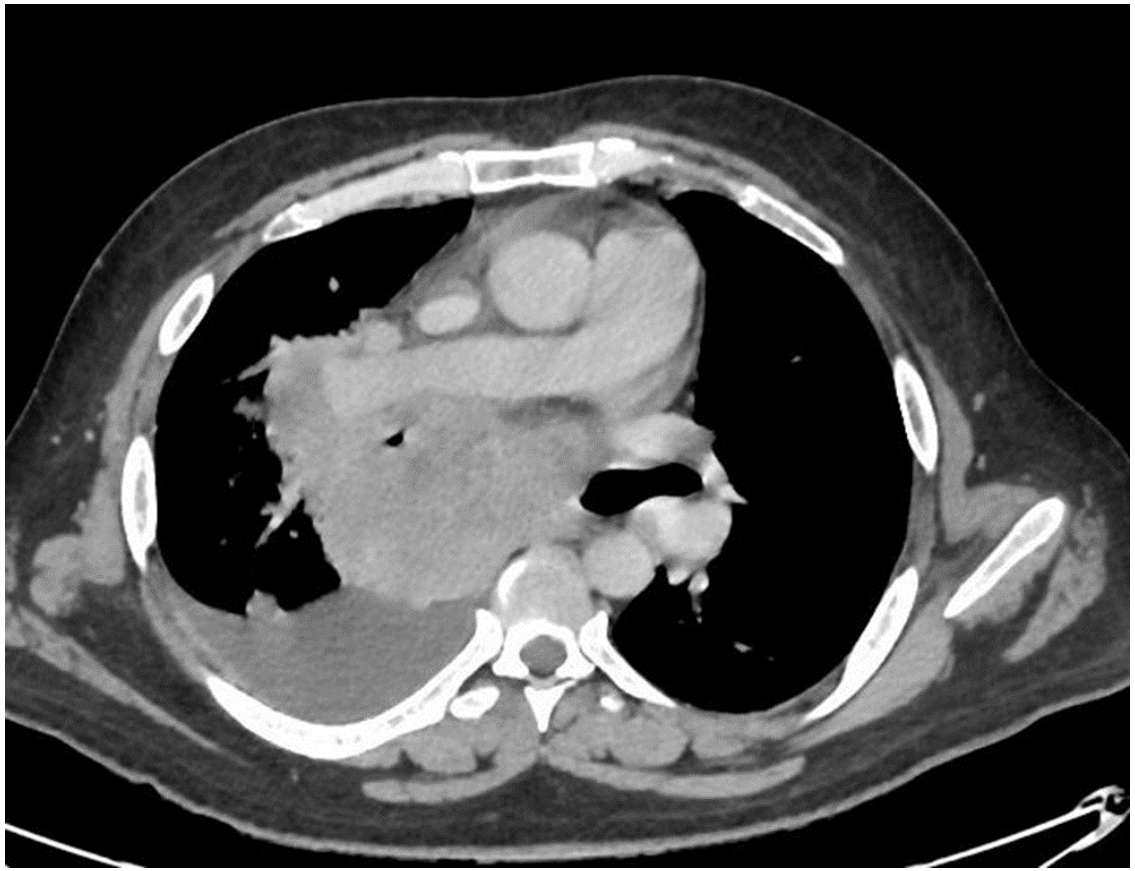
A. Limited disease : is confined to one lung with ipsilateral lymph node metastasis.

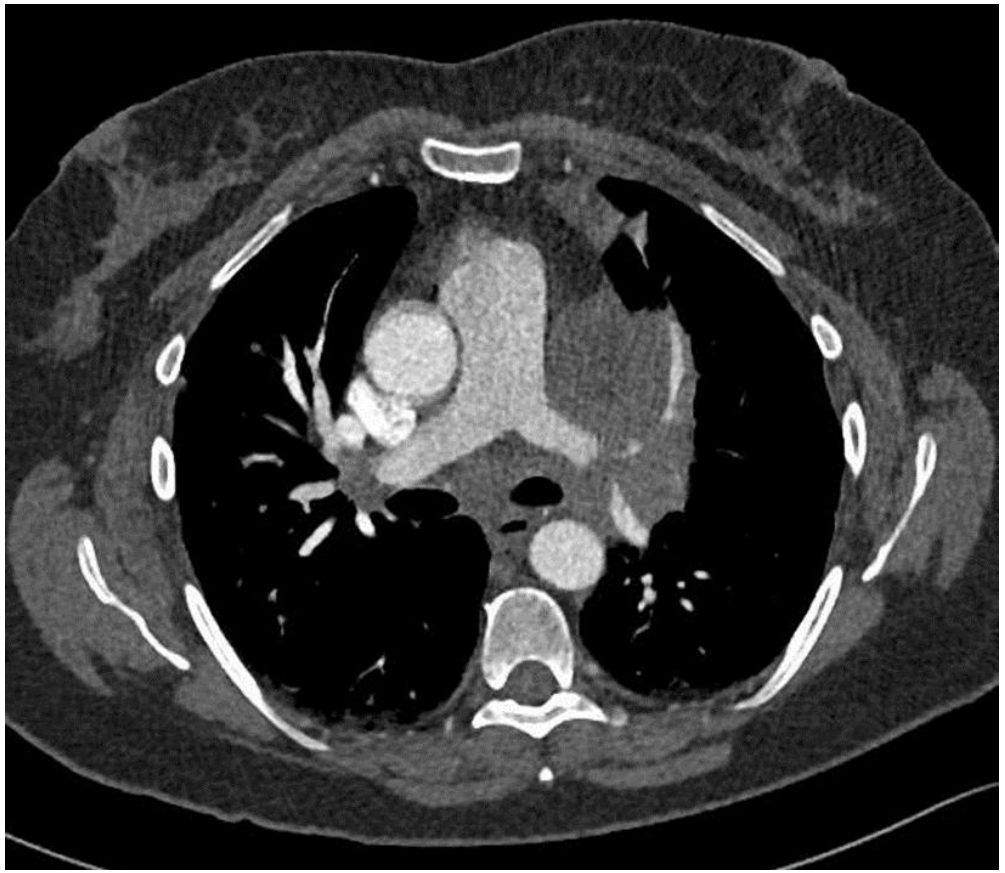
B. Extensive disease: denotes metastases to the contralateral lung and nodes or distant organs (including the pleura).

Small Cell Lung Cancer



Small Cell Lung Cancer





Pancoast Tumor (Superior Sulcus Tumor)

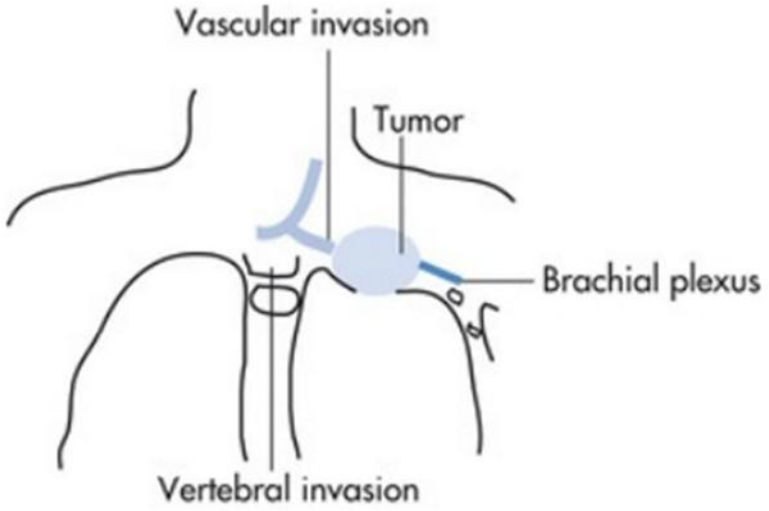
- Tumor located in the lung apex that has extended into the adjacent chest wall.
- Histologically, pancoast tumors are often SCCs.

Clinical Findings

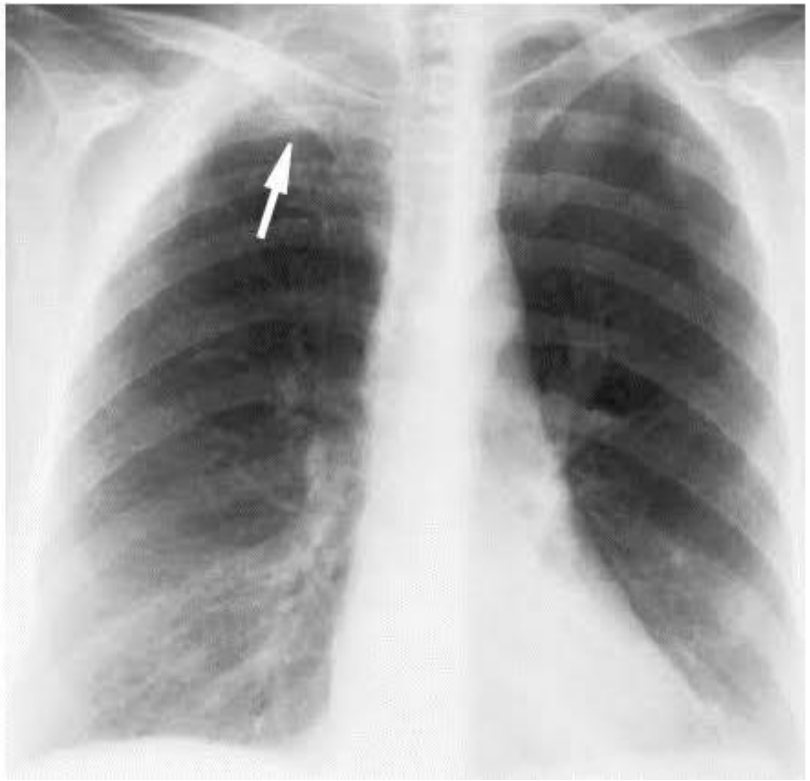
- Horner syndrome
- Pain radiating into arm (invasion of pleura, bone, brachial plexus, or subclavian vessels).

Imgaing Findings

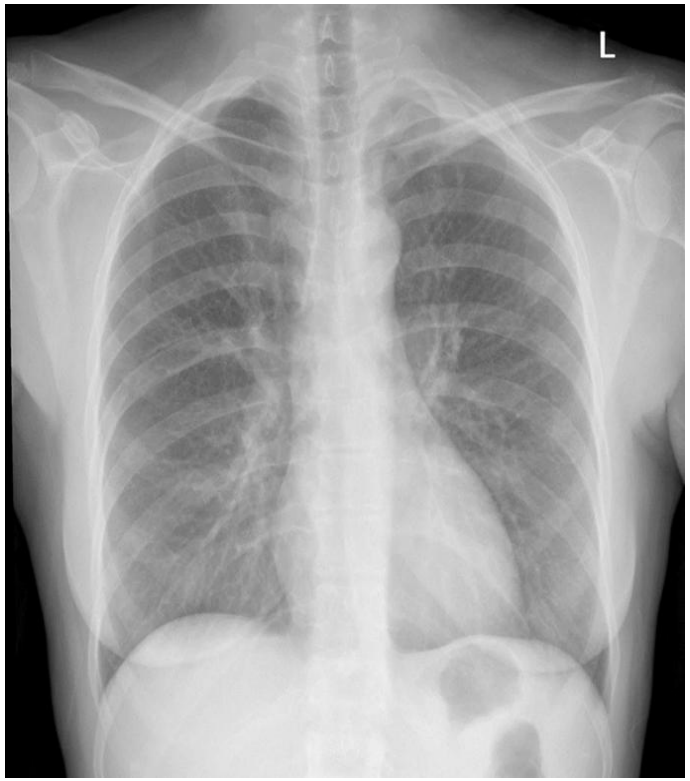
- Apical mass
- Chest wall invasion
- Involvement of subclavian vessels
- Brachial plexus involvement
- Bone involvement: rib, vertebral body



Pancoast Tumor (Superior Sulcus Tumor)







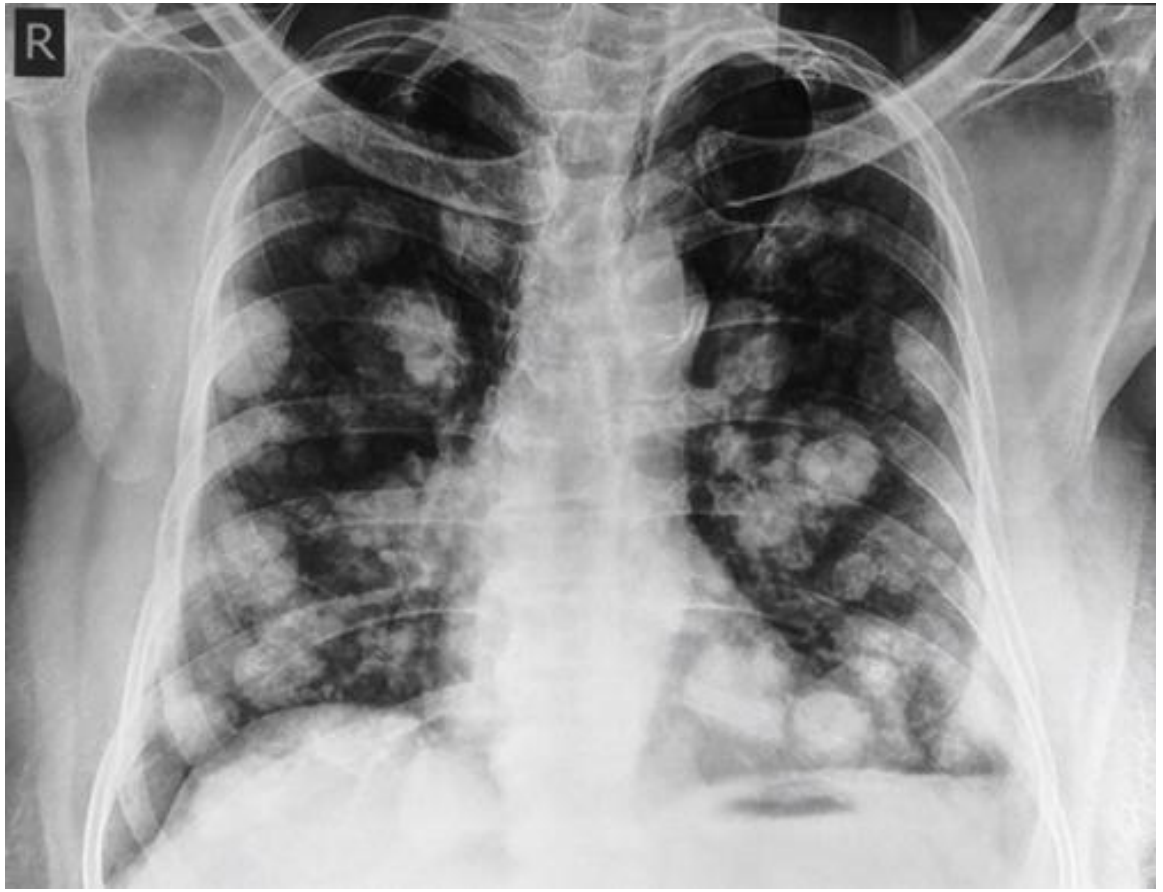
Pancoast Tumor (Superior Sulcus Tumor)



Lung Metastasis

Radiographic Features

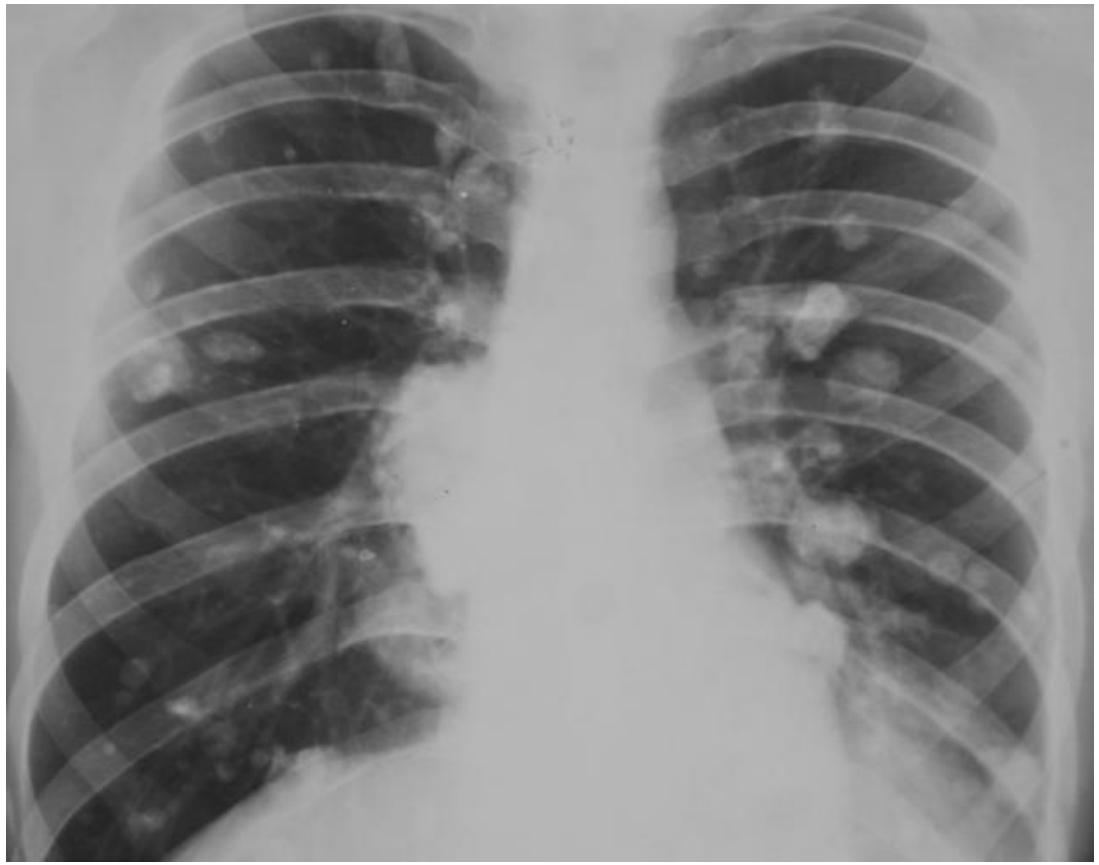
- Multiple lesions, 95% > solitary lesion, 5%
- Lung bases > apices (related to blood flow)
- Peripheral, 90% > central, 10%
- Metastases typically have sharp margins.
- Fuzzy margins can result from peritumoral hemorrhage (like choriocarcinoma)
- Cavitations are common in Squamous Cell CAs from head and neck primary lesions.



Lung Metastasis

Calcified metastases are observed in:

- Bone tumor metastases: Osteosarcoma and Chondrosarcoma
- Mucinous tumors: usually from Ovarian, Colon and Stomach primaries.



References

Keep the source list consistent and academically formatted.

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QUESTIONS & DISCUSSION

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